

Vulnerability of urban street networks to floods

A case study in São Paulo, Brazil

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Pedro Augusto Borges dos Santos

1 Introduction

Objectives



- Investigate the vulnerability of urban street network based on topology and the occurrence of floods.
- Explore the relationship between rainfall, river stage, and flood occurrence, as well as the relationship between topological properties and flood occurrence in the studied area.
- Identify the most vulnerable street sections.

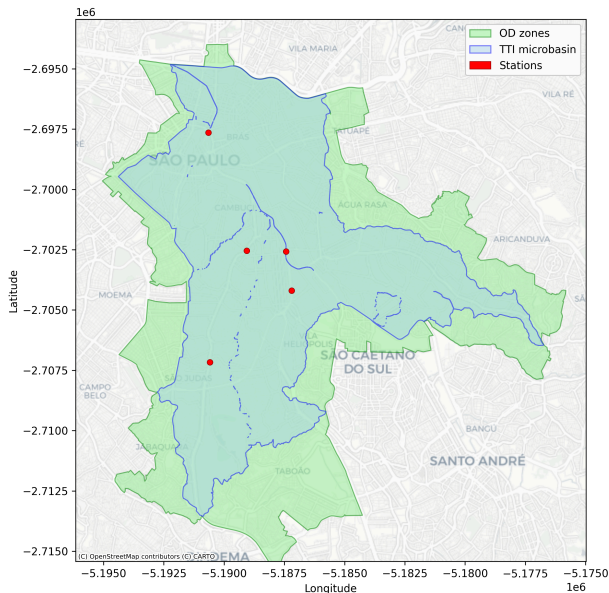
Context



- Resilience of street networks: its capacity to resist, absorb, adapt and transform in the face of environmental impacts (Sharifi, 2019).
- Studying and analyzing the vulnerability of street networks helps in prioritizing planning, budgeting, and maintenance, as well as preparing effective emergency response plans (Balijepalli & Oppong, 2014).

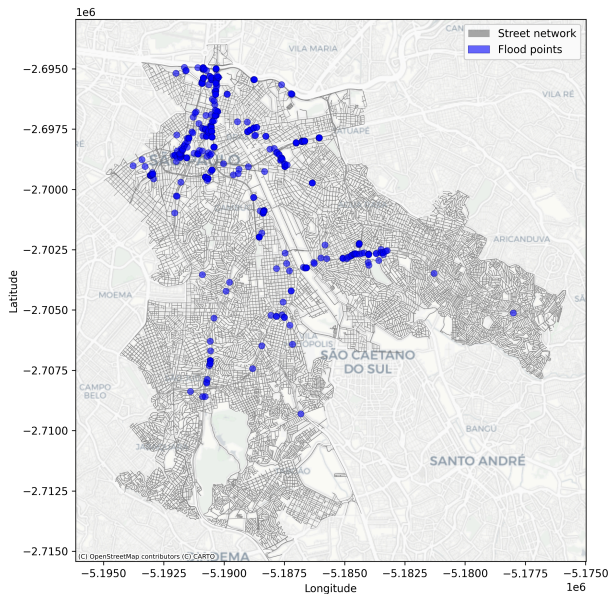
2 Material and methods

Study area and data collection stations



- Tamanduateí (TTI) river basin inside the city of São Paulo
- Origin/destination (OD) zones that intersected the TTI river microbasins
- Five data collection stations, with rainfall and river stage data every 10 minutes, between 2022-01 and 2025-04

Flood and network data



- Street network from 2025-11 (Boeing, 2025)
- Flood points between 2022-01 and 2025-04

Graph topology



Calculation of:

- Node degree

$$k_i = \sum_j a_{ij}$$

- Node clustering coefficient

$$c_i = \frac{2L_i}{k_i(k_i - 1)}$$

- Edge betweenness centrality (EBC) (vulnerability proxy)

$$c_{B(e)} = \sum_{s,t \in V} \frac{\sigma(s, t \mid e)}{\sigma(s, t)}$$

Floods x rainfall x river stage



- Rainfall processing:
 - Calculation of daily values (sum)
 - spearman correlation between stations
 - mean daily value between stations,
 - distribution between flood x non-flood days
- River stage processing:
 - Calculation of daily values (mean)
 - Spearman correlation between stations
 - Mean normalized daily values between stations
 - distribution between flood x non-flood days

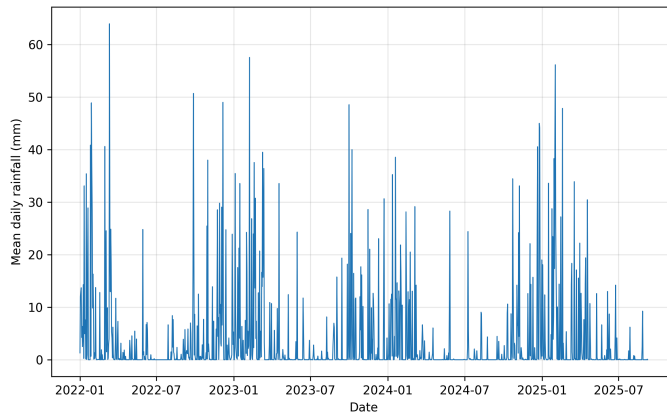
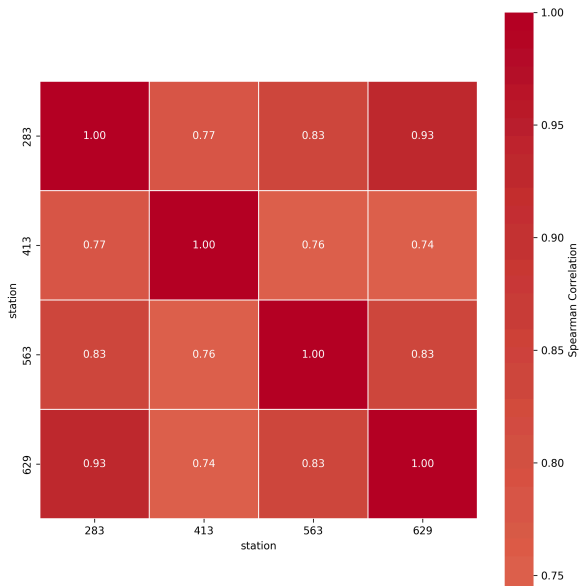
Floods x Vulnerability



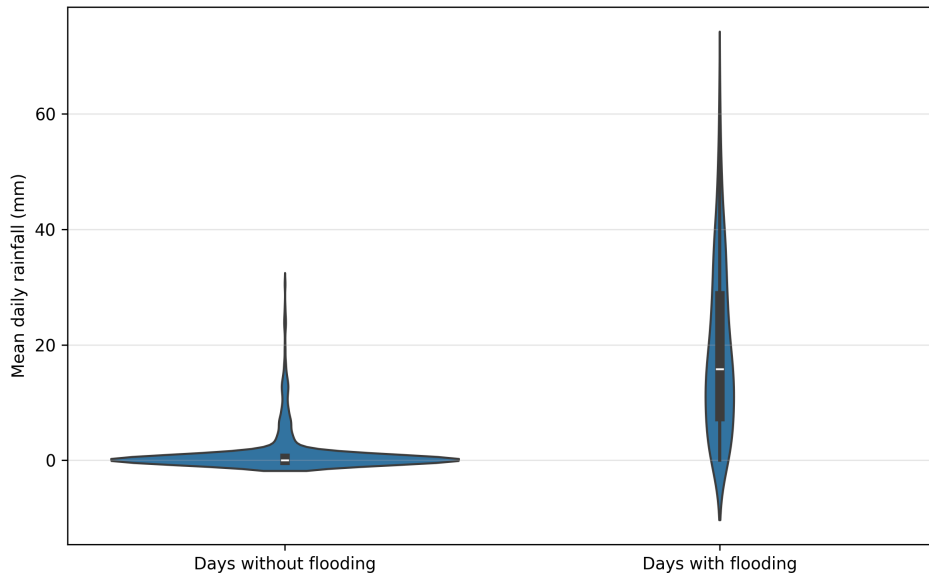
- Comparison between EBC values and flood counts per edge (spatial proximity join)
- Highlight of higher values cases - values above the median, for EBC and flood counts values

3 Preliminary results

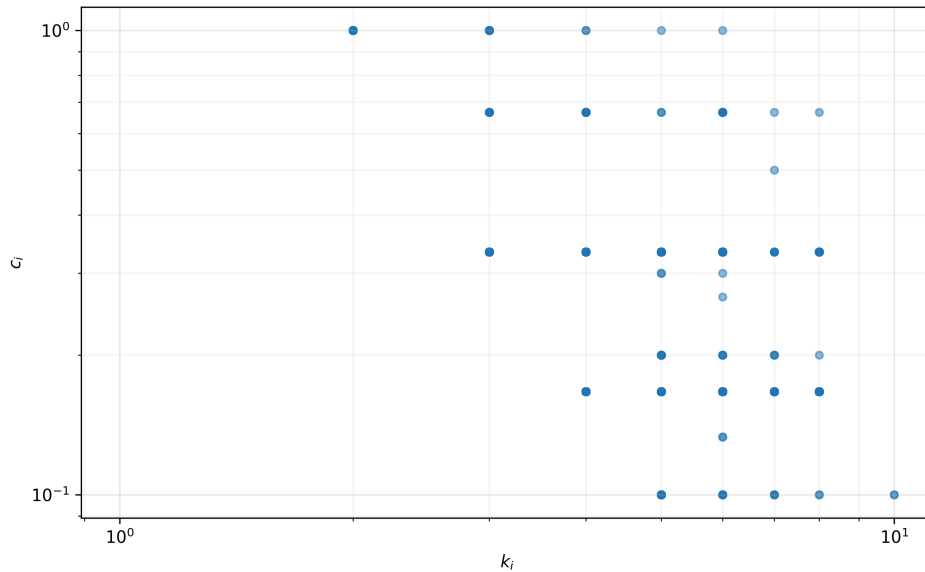
Rainfall



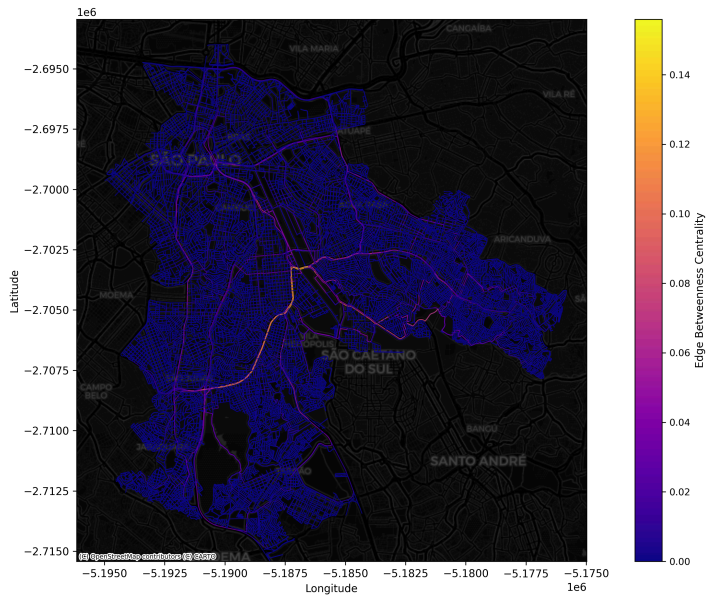
Rainfall x river level



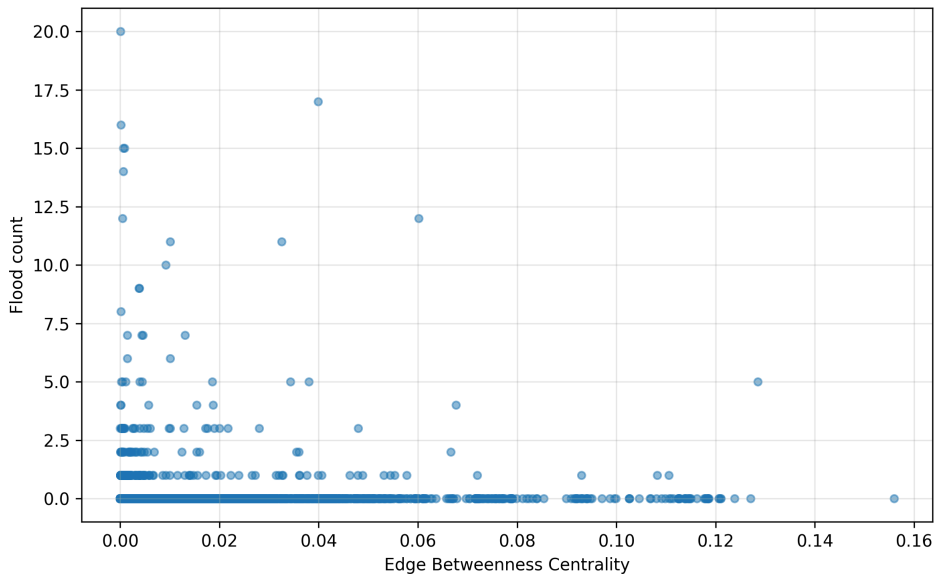
Node degree and clustering



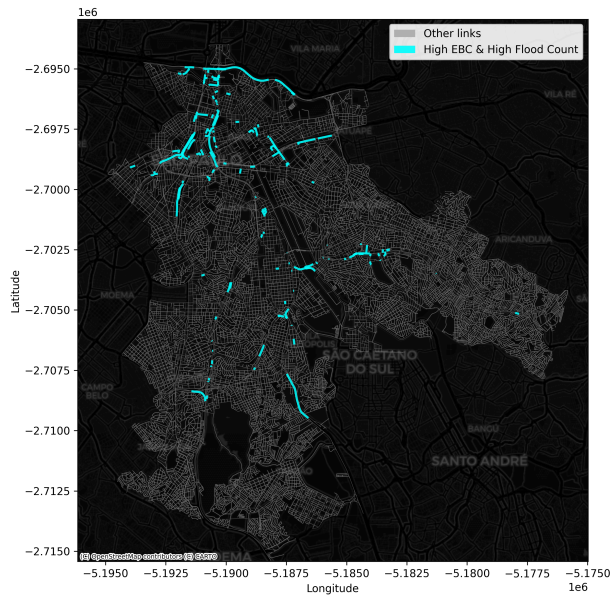
Edge betweenness centrality



EBC x Floods



Vulnerability highlight



4 Bibliography

Bibliography



- Balijepalli, C., & Oppong, O. (2014). Measuring Vulnerability of Road Network Considering the Extent of Serviceability of Critical Road Links in Urban Areas. *Journal of Transport Geography*, 39, 145–155. <https://doi.org/10.1016/j.jtrangeo.2014.06.025>
- Boeing, G. (2025). *Modeling and Analyzing Urban Networks and Amenities with OSMnx*. <https://doi.org/10.48550/ARXIV.2505.00736>
- Sharifi, A. (2019). Resilient Urban Forms: A Review of Literature on Streets and Street Networks. *Building and Environment*, 147, 171–187. <https://doi.org/10.1016/j.buildenv.2018.09.040>